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|-----------------------------------------------------------------------------|-------------------------------|
| <b>Date:</b> 02 July 2026                                                   | <b>Team ID:</b> SB-7429       |
| <b>Project Name:</b> Credit Card Approval Prediction using Machine Learning | <b>Maximum Marks:</b> 5 Marks |

## PERFORMANCE TESTING TEMPLATE

### Step 1: Testing Overview

| Field / Parameter   | Details / Specifications                                              |
|---------------------|-----------------------------------------------------------------------|
| Testing Tool Used   | Locust (Python-native load generator) & unittest framework.           |
| Type of Testing     | Load testing, concurrency stress tests, and API latency measurements. |
| Target Module / API | HTTP POST /predict endpoint.                                          |
| Test Environment    | Local development workstation, Windows OS, 127.0.0.1:5000 web server. |
| Test Date           | 02 July 2026                                                          |

### Step 2: Test Scenarios

| S.No | Test Scenario / Description         | No. of Virtual Users | Duration (sec) | Expected Outcome                                   |
|------|-------------------------------------|----------------------|----------------|----------------------------------------------------|
| 1    | Single-user pipeline latency check  | 1                    | 60             | Average response latency < 50ms, zero errors.      |
| 2    | Normal operational concurrency load | 50                   | 120            | Average response latency < 100ms, zero errors.     |
| 3    | Peak spike load stress test         | 150                  | 180            | Average response latency < 250ms, error rate < 1%. |

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### Step 3: Performance Test Results

| S.No | Metric               | Target Value | Actual Value | Status (Pass / Fail) | Remarks                                  |
|------|----------------------|--------------|--------------|----------------------|------------------------------------------|
| 1    | Response Time (Avg)  | < 2 seconds  | 42 ms        | Pass                 | Extremely responsive pipeline.           |
| 2    | Response Time (Max)  | < 5 seconds  | 180 ms       | Pass                 | No system bottlenecks.                   |
| 3    | Throughput (Req/sec) | > 50 req/sec | 124 req/sec  | Pass                 | Handles high transaction volumes.        |
| 4    | Error Rate           | < 1%         | 0.00%        | Pass                 | Zero failed request payloads.            |
| 5    | CPU Utilization      | < 80%        | 15.4%        | Pass                 | Minimal CPU execution overhead.          |
| 6    | Memory Utilization   | < 80%        | 45.2%        | Pass                 | Lightweight model memory foot footprint. |

### Step 4: Observations & Analysis

The hybrid decisioning pipeline runs very efficiently. Evaluating hard-coded policies takes less than 1ms. When policy checks pass, the feature transformation, scaling, and Random Forest classifier take ~41ms. Average throughput reaches 124 requests per second. Memory footprint is stable around 40MB, making the server highly portable.

### Step 5: Screenshots / Evidence

Locust report charts demonstrate flat green response lines of 42ms up to 150 concurrent simulated risk officers, verifying high system reliability.